

Q1. What is the concept of a weak entity used in data modelling? Define owner entity type and identifying relationship type.

A **weak entity** is an entity that **cannot be uniquely identified by its own attributes** and depends on another entity for its existence.

Key Points:

- A weak entity **does not have a primary key** of its own.
- It is identified using a **partial key** along with the primary key of another entity.
- Represented using a **double rectangle** in ER diagram.

Owner Entity Type:

- The **owner entity** is a strong entity on which the weak entity depends.
- It provides the primary key required to identify the weak entity uniquely.
- Example: *Employee* is owner entity for *Dependent*.

Identifying Relationship Type:

- The relationship that connects a weak entity to its owner entity.
- Represented using a **double diamond**.
- It ensures the existence of the weak entity.

Example:

Dependent (weak entity) depends on Employee (owner entity).

Dependent is identified by (*EmpID + DependentName*).

Q2. Explain the three categories of Data Models.

A **data model** defines how data is structured, stored, and manipulated in a database.

Three Categories of Data Models:

1. Conceptual Data Model

- High-level representation of data.
- Focuses on **what data is stored**, not how.
- Uses **ER diagrams**.
- Used by database designers.

2. Logical Data Model

- Represents data logically without physical storage details.
- Defines tables, attributes, keys, and relationships.
- Example: Relational model.

3. Physical Data Model

- Describes how data is stored physically on disk.
- Includes indexing, storage structure, file organization.
- Used by DBAs.

Q3. Write SQL queries for the given relation schema.

Relations:

employee(ID, person_name, street, city)

works(ID, company_name, salary)

company(company_name, city)

manages(ID, manager_id)

(i) Find employees whose name starts with 'C'

SELECT *

FROM employee

WHERE person_name LIKE 'C%';

(ii) Find ID, name and city of employees who work for "First Bank Corporation" and earn more than Rs. 50000

SELECT e.ID, e.person_name, e.city

FROM employee e, works w

WHERE e.ID = w.ID

AND w.company_name = 'First Bank Corporation'

AND w.salary > 50000;

Q4. Explain Union, Intersection and Set Difference in Relational Algebra with example.

Relational algebra supports **set operations** similar to mathematical sets.

Union (U)

- Combines tuples from two relations.
- Both relations must be **union compatible**.

Example:

$R \cup S \rightarrow$ tuples present in R or S.

Intersection ($\cap$)

- Returns common tuples from both relations.

Example:

$R \cap S \rightarrow$ tuples present in both R and S.

Set Difference ($-$)

- Returns tuples present in first relation but not in second.

Example:

$R - S \rightarrow$ tuples in R but not in S.

Q5. Illustrate the concepts of Schema and Instance using a neat diagram.

Schema:

- The **design or structure** of the database.
- Does not change frequently.
- Defines tables, attributes, and relationships.

Example:

STUDENT(rollNo, name, dept)

Instance:

- The **actual data stored** in the database at a particular time.
- Changes whenever data is inserted, deleted, or updated.

Example Instance:

rollNo name dept

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Schema = Structure

Instance = Data at a moment

Q6. Distinguish between Primary Key, Foreign Key, Super Key and Candidate Key with example.

Key Type	Description
Primary Key	Uniquely identifies a tuple; cannot be NULL
Foreign Key	Refers to primary key of another table
Super Key	Set of attributes that uniquely identifies a tuple
Candidate Key	Minimal super key

Example:

STUDENT(rollNo, email, name)

- Super Key: {rollNo, email}
- Candidate Key: rollNo
- Primary Key: rollNo
- Foreign Key: deptId (from DEPARTMENT table)

Q7. Illustrate the concept of Trigger in SQL with an example.

A **trigger** is a database object that **automatically executes** in response to certain events like INSERT, UPDATE or DELETE.

Uses:

- Enforce integrity constraints
- Maintain consistency
- Automatic validation

Example:

Ensure account balance never becomes negative.

```
CREATE TRIGGER check_balance
```

```
BEFORE UPDATE ON account
```

```
FOR EACH ROW
```

```
BEGIN
```

```
IF NEW.balance < 0 THEN
```

```
SIGNAL SQLSTATE '45000'  
  
SET MESSAGE_TEXT = 'Balance cannot be negative';  
  
END IF;  
  
END;
```

Q8. Write relational algebra expressions for the following.

Relations:

Student(rollNo, name, dept, year)

Course(courseId, title, dept)

Enroll(rollNo, courseId, grade)

(a) Names of students enrolled in Computer Science department

$\pi \text{ name } (\sigma \text{ dept} = \text{'Computer Science'} (\text{Student}))$

(b) Students enrolled in more than one course

$\pi \text{ rollNo } (\text{Enroll}) - \pi \text{ rollNo } (\sigma \text{ courseId} = \text{NULL} (\text{Enroll}))$

Q9. Explain the concept of a weak entity used in data modelling. Define owner entity type and identifying relationship type.

A **weak entity** is an entity that **cannot be uniquely identified by its own attributes** and depends on a strong entity for identification.

Characteristics:

- Does not have a primary key of its own
- Uses a **partial key** along with owner entity's key
- Represented by **double rectangle** in ER diagram

Owner Entity Type:

- The strong entity on which the weak entity depends
- Provides the primary key
- Example: EMPLOYEE is owner of DEPENDENT

Identifying Relationship Type:

- Relationship that links weak entity with owner entity
- Represented by **double diamond**

- Ensures existence dependency

Q10. Differentiate structured, unstructured and semi-structured data with examples.

Type	Description	Example
Structured Data	Organized in rows and columns	Tables in DBMS
Semi-structured Data	Partial structure with tags	XML, JSON
Unstructured Data	No fixed format	Images, videos, emails

Key Point:

DBMS mainly handles **structured and semi-structured data** efficiently.

Q11. Write briefly about any two relational database integrity constraints.

1. Entity Integrity Constraint

- Primary key **cannot be NULL**
- Ensures unique identification of tuples

2. Referential Integrity Constraint

- Foreign key must match a primary key in referenced table
- Prevents invalid references

Example:

Employee(deptId) → Department(deptId)

Q12. Analyse the difference between theta join and natural join operations.

Theta Join	Natural Join
Uses any comparison operator	Uses equality only
Condition explicitly specified	Automatically matches common attributes
Attribute duplication exists	Duplicate attributes removed

Notation:

Theta Join: $R \bowtie_{\theta} S$

Natural Join: $R \bowtie S$

Q13. (a) Difference between logical and physical data independence. Which is harder to achieve? Why?

Logical Data Independence:

- Change in logical schema does not affect external views
- Example: Adding new column to table

Physical Data Independence:

- Change in physical storage does not affect logical schema
- Example: Changing file structure

Harder to Achieve:

Logical data independence is harder because application programs depend on logical structure.

Q13. (b) Interpret the given ER diagram.

Interpretation:

- **Artist** entity compiles **Album**
 - Each album contains multiple **Tracks**
 - Tracks are played and stored in **Played** entity
 - One-to-many and many-to-one relationships are shown
 - Attributes like artist_id, album_id, track_id uniquely identify entities
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Q14. (a) Differentiate between two-tier and three-tier client-server architecture.

Two-Tier

Three-Tier

Client directly communicates with DB Middle layer present

Less secure

More secure

Used in small applications

Used in web applications

Layers in Three-Tier:

1. Presentation layer

2. Application layer
3. Database layer

Q14. (b) Draw ER diagram based on given information (Manufacturers, Products, Customers, Orders).

Entities Identified:

- Manufacturer(name, address, phone)
- Product(model_no, type)
- Customer(SSN, email, address)
- Order(order_no, date)

Relationships:

- Manufacturer **produces** Product (1:N)
- Customer **places** Order (1:N)
- Order **contains** Product (M:N with quantity)

Q15. (a) Write SQL DDL statements for given schema.

Create Tables:

```
CREATE TABLE Company(  
  comp_id INT PRIMARY KEY,  
  cname VARCHAR(20),  
  address VARCHAR(30),  
  turnover INT  
);
```

```
CREATE TABLE Employee(  
  eid INT PRIMARY KEY,  
  name VARCHAR(20),  
  designation VARCHAR(20),
```

salary INT,
comp_id INT,
FOREIGN KEY(comp_id) REFERENCES Company(comp_id)
);

Insertion Order:

1. Company
 2. Employee
-

Q15. (b) Answer the following.

(i) DROP vs DELETE

- DROP removes table structure
- DELETE removes records only

(ii) ALTER vs UPDATE

- ALTER modifies table structure
- UPDATE modifies data

Foreign Key Constraint:

- Maintains referential integrity
 - Prevents invalid references
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Q16. Write relational algebra expressions for UNIVERSITY database.

(i) Department name and HOD details

π name, hod, sex, phone (DEPARTMENT $\bowtie$ PROFESSOR)

(ii) Courses offered by each department

π cname, deptNo (COURSE)

(iii) Students registered for all CS courses

π rollNo (ENROLLMENT $\div$ π courseId (σ deptNo='CS' (COURSE)))

Q17. (a) Explain Three Schema Architecture with diagram.

The **Three Schema Architecture** (ANSI/SPARC architecture) separates the database system into three levels to achieve data abstraction and data independence.

Levels of Architecture:

1. Physical Level

- Lowest level
- Describes how data is stored on disk
- Deals with files, indexes, storage blocks

2. Logical Level

- Middle level
- Describes what data is stored and relationships
- Used by programmers and DBAs

3. View Level

- Highest level
- Describes user views of data
- Hides complexity from users

Advantage:

Provides **data abstraction** and **data independence**.

Q17. (b) List any six advantages of DBMS over File System.

1. Reduced data redundancy
 2. Improved data consistency
 3. Better data security
 4. Data sharing among users
 5. Backup and recovery support
 6. Concurrent access support
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Q18. (a) An organization plans to modify its database structure without affecting application programs. Which data independence is required? Explain.

The organization requires **Logical Data Independence**.

Explanation:

- Logical data independence allows changes in logical schema
- Does not affect application programs or user views

Example:

- Adding a new column to a table
- Splitting a table into two tables

Logical data independence is harder to achieve compared to physical data independence.

Q18. (b) List the database languages supported by a DBMS and mention their purpose.

Database Languages:**Language Purpose**

DDL	Defines database structure
DML	Manipulates data
DCL	Controls user access
TCL	Manages transactions

Examples:

CREATE (DDL), SELECT (DML), GRANT (DCL), COMMIT (TCL)

Q19. (a) ER modeling for given sales scenario.

(i) Entities and Relationships:**Entities:**

- Salesperson (SSN, name)
- Customer (CustID, name)
- Order (OrderID, date)
- Item (ItemID, name)
- Part (PartID, name)
- Supplier (SupplierID, name)

- Employee (EmpID, name)

Relationships:

- Salesperson manages Salesperson (1:N)
- Salesperson handles Customer (1:N)
- Customer places Order (1:N)
- Order contains Item (M:N)
- Item assembled from Part (M:N)
- Supplier supplies Part (M:N)

Q19. (b) Write SQL queries for given schema.

(i) Employees whose name starts with 'C'

```
SELECT * FROM employee
WHERE person_name LIKE 'C%';
```

(ii) Name of managers of each company

```
SELECT DISTINCT e.person_name
FROM employee e, manages m
WHERE e.ID = m.manager_id;
```

(iii) Employees of First Bank Corporation earning > 50000

```
SELECT e.ID, e.person_name, e.city
FROM employee e, works w
WHERE e.ID = w.ID
AND w.company_name='First Bank Corporation'
AND w.salary > 50000;
```

(iv) Companies with higher average salary than First Bank Corporation

```
SELECT company_name
FROM works
GROUP BY company_name
HAVING AVG(salary) >
```

```
(SELECT AVG(salary) FROM works  
WHERE company_name='First Bank Corporation');
```

Q20. (a) Write SQL queries using aggregate functions.

Relation:

EMPLOYEE(Emp_ID, Emp_Name, Department, Salary)

(i) Total salary

```
SELECT SUM(Salary) FROM EMPLOYEE;
```

(ii) Average salary

```
SELECT AVG(Salary) FROM EMPLOYEE;
```

(iii) Maximum salary

```
SELECT MAX(Salary) FROM EMPLOYEE;
```

Q20. (b) Define a trigger and write a trigger to prevent negative balance.

A **trigger** is a database object that automatically executes when a specified event occurs.

Trigger Example:

```
CREATE TRIGGER prevent_negative_balance  
BEFORE UPDATE ON account  
FOR EACH ROW  
BEGIN  
IF NEW.balance < 0 THEN  
SIGNAL SQLSTATE '45000'  
SET MESSAGE_TEXT = 'Balance cannot be negative';  
END IF;  
END;
```

Integrity:

Ensures account balance never becomes negative.

Q21. (a) Explain Three Schema Architecture and differentiate data independence.

Physical Data Independence:

- Changes in physical storage
- Does not affect logical schema

Logical Data Independence:

- Changes in logical schema
- Does not affect user views

Comparison:

Logical data independence is harder to achieve.

Q21. (b) List strong entity types in the given BANK ER diagram.

Strong Entities:

- BANK
 - BANK_BRANCH
 - ACCOUNT
 - LOAN
 - CUSTOMER
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Q22. Distinguish between two-tier and three-tier architecture.

Two-Tier

Three-Tier

Client directly accesses DB Application server in between

Less scalable

Highly scalable

Less secure

More secure

Q23. Write SQL queries for given COMPANY database.

(i) Employees working in 'Research' department

SELECT Fname, Address

FROM EMPLOYEE e, DEPARTMENT d

WHERE e.Dno = d.Dnumber

AND d.Dname='Research';

(ii) Employee and supervisor name

SELECT e.Fname, s.Fname

FROM EMPLOYEE e, EMPLOYEE s

WHERE e.SuperSSN = s.SSN;

(iii) Employees working on all projects of department 5

SELECT ESSN

FROM WORKS_ON

WHERE Pno IN

(SELECT Pnumber FROM PROJECT WHERE Dnum=5);

Q24. Write relational algebra expressions.

(i) Students registered for all CS courses

$\pi \text{ rollNo } (\text{ENROLLMENT} \div \pi \text{ courseId } (\sigma \text{ deptNo}='CS' (\text{COURSE})))$

(ii) Departments with no lady professor

$\pi \text{ deptId } (\text{DEPARTMENT}) - \pi \text{ deptNo } (\sigma \text{ sex}='F' (\text{PROFESSOR}))$

(iii) RollNo of girl students with at least one S grade

$\pi \text{ rollNo } (\sigma \text{ sex}='F' \text{ AND } \text{grade}='S' (\text{STUDENT} \bowtie \text{ENROLLMENT}))$